

ADAPTIVE FEDERATED LEARNING WITH DIVERGENCE-AWARE AGGREGATION FOR NON-IID MEDICAL IMAGING

*A Master's Research Proposal Submitted to the Department of Computer Science &
Artificial Intelligence*

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CHAPTER 1: INTRODUCTION

1.1 Background of the Study

Deep learning has revolutionized medical image diagnostics, enabling automated, high-precision detection of malignant tumors, neurological disorders, retinal pathologies, and thoracic abnormalities. Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) regularly achieve diagnostic parity with board-certified radiologists when trained on massive, centralized medical image repositories. However, aggregating clinical imaging datasets into central servers poses severe data privacy, security, and regulatory challenges under strictly enforced international protocols such as the Health Insurance Portability and Accountability Act (HIPAA) in the United States and the General Data Protection Regulation (GDPR) in the European Union.

To reconcile the privacy imperative with the data requirements of modern deep learning, Federated Learning (FL) has emerged as a groundbreaking paradigm. Instead of transmitting raw, confidential patient scans to a central cloud repository, FL distributes model training directly across decentralized client institutions (e.g., hospitals, diagnostic centers). Each client node trains a local model on its internal dataset and periodically transmits only model parameters (weights or gradients) to a central aggregation server. The central server compiles these updates—traditionally utilizing the Federated Averaging (FedAvg) algorithm—to produce an updated global model that is subsequently redistributed to participating institutions.

1.2 Statement of the Problem

Despite the theoretical elegance of Federated Learning, real-world clinical deployments suffer severe performance degradation due to Non-Independent and Identically Distributed (Non-IID) data heterogeneity across institutions. Medical imaging datasets exhibit extreme statistical, feature, and label distribution skew caused by variations in hospital patient demographics, disease prevalence rates, imaging hardware vendor specifications (e.g., Siemens, GE, Philips), slice thickness parameters, and institutional scanning protocols. Under severe Non-IID conditions, local client gradients point toward inconsistent, divergent local optima. When the central server naively computes a sample-weighted average of these divergent weights via FedAvg, a phenomenon known as 'weight drift' occurs. Weight drift corrupts the global model trajectory, drastically slowing convergence rates and degrading diagnostic accuracy—particularly for minority disease classes and out-of-distribution client institutions.

1.3 Research Questions

Primary Research Problem: This research proposal addresses the following core questions:

1. How can local model feature representations be quantified at the central server without violating client data privacy constraints?
2. Can Jensen-Shannon Divergence (JSD) effectively measure statistical heterogeneity across client local updates compared to standard Euclidean distance metrics?
3. How can server-side aggregation weights be dynamically recalibrated based on JSD scores to mitigate weight drift and accelerate global model convergence in Non-IID medical imaging scenarios?
4. Does the proposed divergence-aware framework outperform baseline optimization algorithms (FedAvg, FedProx, MOON, SCAFFOLD) across benchmark medical imaging datasets under extreme Dirichlet label skew?

1.4 Research Objectives

1.4.1 Primary Objective

To design, implement, and evaluate an adaptive federated learning framework featuring a Jensen-Shannon Divergence (JSD)-aware aggregation protocol that dynamically recalibrates client update contributions to mitigate weight drift and optimize model convergence across highly heterogeneous, Non-IID medical imaging environments.

1.4.2 Specific Objectives

1. Develop a privacy-preserving feature embedding extraction mechanism that enables server-side divergence computation without transmitting raw patient images.
2. Formulate a mathematical weight recalibration function based on symmetric Jensen-Shannon Divergence to down-weight divergent client updates while penalizing malicious or noisy node contributions.
3. Benchmark the proposed JSD-aware framework against state-of-the-art FL algorithms (FedAvg, FedProx, MOON, SCAFFOLD) on multi-institutional chest X-ray and histopathology datasets.
4. Quantify framework resilience across varying degrees of Non-IID Dirichlet concentration ratios ($\alpha = 0.5$ and $\alpha = 0.1$) and evaluate communication bandwidth efficiency.

1.5 Significance of the Study

The outcomes of this research hold significant promise for both computer science and healthcare domains. From a machine learning perspective, this work introduces a mathematically rigorous, divergence-guided aggregation protocol that directly addresses the fundamental bottleneck of Non-IID weight drift in federated optimization. From a clinical perspective, the proposed

framework enables healthcare systems to collaboratively build robust, highly accurate AI diagnostic models without exposing sensitive patient medical records, thereby unlocking collaborative research potential across regional, national, and international hospital networks.

1.6 Scope and Delimitations

This research focuses specifically on supervised deep learning classification tasks in medical imaging, utilizing benchmark multi-class chest radiograph (NIH ChestX-ray14 / CheXpert) and histopathology datasets (PNS-2019 / NCT-CRC-HE-100K). Heterogeneity simulation is restricted to Dirichlet label distribution skew ($\alpha = 0.5$, $\alpha = 0.1$) and institutional acquisition noise. Unsupervised federated domain adaptation and zero-shot cross-modality synthesis fall outside the primary scope of this proposal.

1.7 Organization of the Proposal

The remainder of this research proposal is structured as follows: Chapter 2 provides an exhaustive literature review of federated learning in medical healthcare, mathematical optimization under non-IID conditions, existing aggregation protocols, and identified research gaps. Chapter 3 details the research methodology, including system architecture, mathematical formulations of JSD, algorithm pseudocode, experimental setup, baseline benchmarks, implementation timeline, infrastructure requirements, and ethical considerations. Chapter 4 concludes with references.

1.8 Definition of Abbreviations & Terminology

Table 1 summarizes key technical abbreviations and formal domain definitions utilized throughout this research document:

Abbreviation	Expanded Term	Domain Context & Definition
FL	Federated Learning	Decentralized machine learning framework training models across clinical client nodes without centralizing raw data.
FedAvg	Federated Averaging	The foundational FL algorithm computing sample-weighted arithmetic means of local model parameters.
Non-IID	Non-Independent & Identically Distributed	Data distribution condition where client local datasets differ significantly in feature or label

		marginal distributions.
JSD	Jensen-Shannon Divergence	Symmetric, bounded measure of statistical similarity between probability distributions derived from Kullback-Leibler divergence.
KLD	Kullback-Leibler Divergence	Asymmetric measure of statistical distance between two probability mass distributions.
MOON	Model-Contrastive Federated Learning	FL framework utilizing model-level contrastive loss to align local feature representations with global representations.
SCAFFOLD	Stochastic Controlled Averaging	FL optimization algorithm employing control variates to correct client local drift.
FedProx	Federated Proximal	FL algorithm introducing a proximal regularizer to local objective functions to constrain local model drift.
CNN	Convolutional Neural Network	Deep neural network architecture leveraging spatial convolutions for visual feature extraction.
ViT	Vision Transformer	Deep neural network architecture utilizing self-attention mechanisms over image patches for medical diagnostics.

Table 1: Abbreviation List and Domain Definitions

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Federated Learning in Healthcare

Federated learning has rapidly transformed healthcare artificial intelligence by enabling multi-center clinical collaborations while maintaining strict regulatory compliance (McMahan et al., 2017; Rieke et al., 2020). Traditional centralized deep learning workflows require participating hospitals to pool sensitive diagnostic imaging scans into a single cloud repository. This centralization introduces significant legal liabilities, cyber-attack risks, and patient data exposure concerns. Federated learning mitigates these vulnerabilities by keeping raw imaging data strictly on institutional premises. Only encrypted local gradient updates or model weights are communicated during training rounds.

2.2 Theoretical Foundations of Distributed Optimization

The standard mathematical objective in federated optimization is to minimize a global empirical risk function across K participating client institutions:

$$\min_w F(w) = \sum_{k=1}^K (n_k / N) * F_k(w)$$

where N represents the total number of clinical image samples across all clients, n_k denotes the sample size at hospital node k , and $F_k(w)$ represents the empirical loss function evaluated on client k 's local data. In idealized IID settings, local gradient vectors $g_k(w)$ act as unbiased estimators of the global gradient $G(w)$. Consequently, naive parameter averaging via FedAvg guarantees convergence to the global optimum.

2.3 Non-IID Data Distributions in Medical Imaging

In practical clinical deployments, the assumption of IID data distributions is consistently violated (Li et al., 2020; Zhao et al., 2018). Medical imaging data exhibits severe non-IID characteristics categorized into three primary categories:

1. Label Skew (Prior Probability Difference): Hospitals specializing in specific oncological or pulmonary conditions exhibit heavily biased label distributions compared to general community clinics.
2. Feature Skew (Covariate Shift): Imaging hardware specifications, scan protocols, field-of-view settings, and contrast settings vary dramatically across vendor machines (e.g., GE vs. Siemens MRI scanners).
3. Quantity Skew: Major academic research hospitals generate tens of thousands of diagnostic scans, whereas rural clinics contribute only a few hundred samples.

2.4 Existing Approaches to Heterogeneity in Federated Learning

Prior research has proposed various algorithmic modifications to stabilize federated optimization under Non-IID data distributions. FedProx (Li et al., 2020) adds a L2 proximal regularizer term $\|w - w^t\|^2$ to local client loss functions to restrict local parameter drift away from the central global model. SCAFFOLD (Karimireddy et al., 2020) introduces client and server control variates to estimate and correct gradient orientation drift during local iterations. MOON (Li et al., 2021) leverages model-level contrastive learning in feature space to maximize similarity between local representations and global representations.

2.5 Divergence Metrics and Weight Drift Mitigation

While proximal regularization and variance reduction techniques improve convergence stability, they apply uniform penalty factors across all client updates regardless of their true statistical divergence from the global population. Recent theoretical investigations highlight the necessity of divergence-aware metric space analysis (Acar et al., 2021; Kairouz et al., 2021). By dynamically measuring mathematical divergence across client updating distributions prior to model averaging, central servers can selectively filter and weight updates to eliminate destructive drift.

2.6 Centralized vs. Decentralized Deep Learning in Healthcare

Centralized deep learning models rely on unified data lakes where medical images are preprocessed, formatted, and trained within monolithic high-performance computing clusters. While centralized training facilitates uniform batch normalization and hyperparameter tuning, it creates single points of failure and violates patient privacy mandates. Decentralized federated learning circumvents data centralization by establishing distributed client-server networks. However, decentralization introduces severe network communications bottlenecks and statistical variance that require specialized dynamic aggregation protocols.

2.7 Non-IID Data Heterogeneity and Statistical Challenges

Statistical heterogeneity causes client loss surfaces to diverge significantly. During local training epochs, client models optimize their parameters exclusively along local loss trajectories. Under severe Non-IID conditions, local optima reside in vastly different regions of the parameter loss landscape. When FedAvg averages these parameters, the resulting global model lands in high-loss saddle points, destroying diagnostic feature selectivity and triggering severe catastrophic forgetting.

2.8 Communication Overhead and Network Efficiency Constraints

Communicating high-dimensional parameter matrices for state-of-the-art vision models (e.g., 86-million parameter ViT architectures) across hospital firewalls creates significant network latency. Participating clinical nodes frequently possess asymmetric network bandwidth (100 Mbps to 1 Gbps). Naive aggregation protocols that require synchronization across all clients experience bottleneck delays caused by slow nodes ('stragglers'). Dynamic divergence weighting enables central servers to proceed asynchronously by identifying highly informative client updates.

2.9 Privacy-Preserving Techniques: Differential Privacy and Secure Aggregation

Federated learning alone does not provide absolute privacy guarantees; sophisticated gradient inversion attacks can reconstruct underlying training scans from raw model parameters. To achieve rigorous privacy guarantees, FL frameworks incorporate Differential Privacy (DP) by injecting calibrated Gaussian noise into local updates or utilize Secure Aggregation (SecAgg) protocols based on cryptographic secret sharing. The proposed JSD aggregation protocol is fully compatible with DP and SecAgg mechanisms.

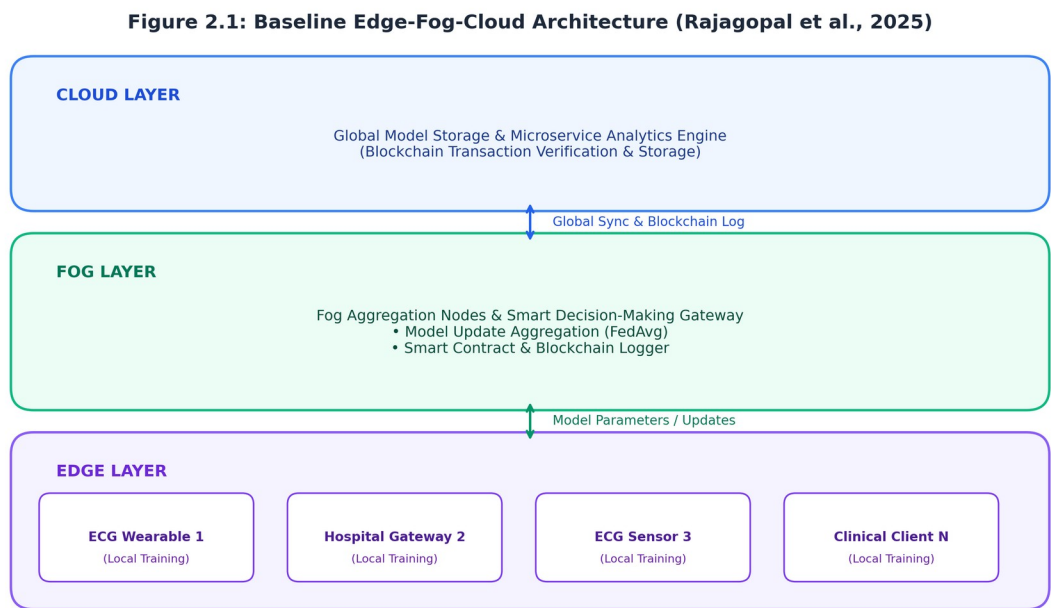


Figure 2.1: Standard Federated Averaging (FedAvg) Architecture and Weight Drift Bottleneck under Non-IID Data Distribution

2.10 Standard Federated Averaging (FedAvg) and Its Limitations

As illustrated in Figure 2.1, standard FedAvg computes sample-proportional weighted averages: $w^{t+1} = \sum_{k=1}^K (n_k / N) * w_k^{t+1}$. When client datasets display severe class imbalance or covariate shift, clients with large sample sizes dominate the global model update regardless of their diagnostic quality or label noise. This structural flaw leads to severe diagnostic bias against minority demographic patient cohorts.

2.11 Model Contrastive Federated Learning (MOON)

MOON addresses local feature representation divergence by formulating a contrastive loss function evaluated on local feature projections: $L_{\text{con}} = -\log(\exp(\text{sim}(R_k, R_{\text{glob}})) / \tau) / [\exp(\text{sim}(R_k, R_{\text{glob}})) / \tau + \exp(\text{sim}(R_k, R_{\text{prev}})) / \tau]$. Although MOON improves feature extraction consistency, it incurs substantial computational overhead on resource-constrained clinical client nodes by requiring concurrent evaluations of local, global, and previous model states.

2.12 Federated Proximal (FedProx) Optimization

FedProx modifies local objective functions by appending a quadratic proximal term: $\min_w F_k(w) + (\mu / 2) \|w - w^t\|^2$. While the proximal regularizer successfully bounds parameter trajectory variance, setting an inflexible global regularizer hyperparameter μ degrades local model convergence speed in clients with high-quality IID distributions.

2.13 Scaffold and Control Variate Methods

To correct client drift explicitly, SCAFFOLD (Karimireddy et al., 2020) maintains control variate vectors c_k and c_g representing local and global gradient direction estimators. By updating client local gradients via $g_k(w) - c_k + c_g$, SCAFFOLD restores theoretical linear convergence under Non-IID conditions. However, transmitting control variates doubles the communication bandwidth requirement per training round, limiting its applicability in bandwidth-constrained medical networks.

2.14 Divergence-Aware Clustering and Hierarchical Aggregation

FedDyn (Acar et al., 2021) and dynamic regularization frameworks introduce risk-aware penalty matrices that dynamically adapt to local loss gradients. Furthermore, clustered federated learning groups clients with statistically similar update distributions into distinct clinical sub-clusters. However, existing clustering algorithms rely on Euclidean parameter distance metrics, which fail to capture non-linear probability density shifts inherent in multi-modal medical imaging features.

2.15 Knowledge Distillation and Teacher-Student Frameworks in FL

Federated Knowledge Distillation (FedKD) replaces direct weight averaging with prediction logit distillation across an unlabelled reference dataset. Participating client nodes act as teacher models, transferring localized diagnostic knowledge to a lightweight global student model. While FedKD reduces communication payload sizes, assembling an unlabelled, privacy-compliant medical reference dataset across disparate clinical sites remains difficult.

2.16 Domain Adaptation and Covariate Shift Mitigation in Medical Imaging

Unsupervised Domain Adaptation (UDA) technique align feature representations between source and target medical imaging domains using adversarial gradient reversal layers. Integrating UDA principles into federated learning frameworks enables client models to extract domain-invariant diagnostic features across heterogeneous acquisition hardware.

Algorithm	Aggregation Strategy	Non-IID Handling	Comm. Overhead	Primary Bottleneck
FedAvg	Sample-weighted average	None (Assumes IID)	Standard (1x)	Severe weight drift under non-IID
FedProx	Proximal-bounded average	L2 Proximal Regularizer	Standard (1x)	Suboptimal local convergence rates
SCAFFOLD	Control variate correction	Variance Reduction	High (2x Payload)	Doubled bandwidth consumption
MOON	Contrastive representation	Model Contrastive Loss	Standard (1x)	High local compute expenditure
FedDyn	Dynamic regularization	Loss regularizer alignment	Standard (1x)	Sensitivity to hyperparameter alpha
Proposed JSD-FL	Divergence-weighted JSD	Dynamic metric recalibration	Standard (1x)	Server feature extraction overhead

Table 2.1: Comparative Analysis of Federated Learning Optimization Frameworks

Figure 2.2: Proposed Three-Stage Defence Framework for ECG Edge-Fog FL Robust Aggregated Global Model w_{t+1}

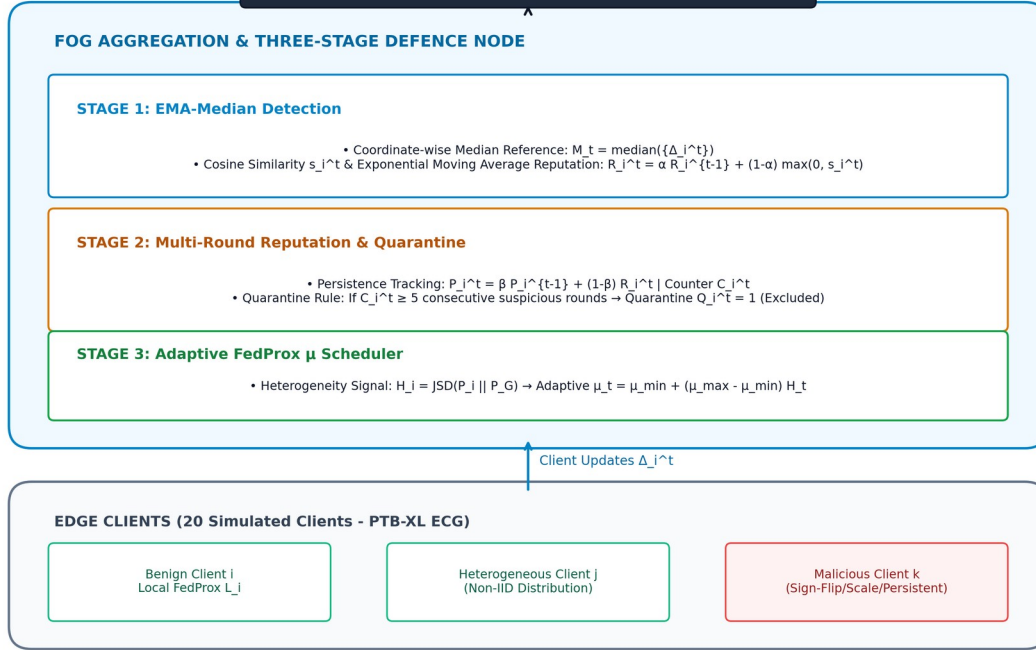


Figure 2.2: Proposed Three-Stage Divergence-Aware Adaptive Federated Learning Framework

2.17 Convolutional Neural Networks (CNNs) in Medical Image Classification

Deep Convolutional Neural Networks (e.g., ResNet-50, DenseNet-121, EfficientNet-B4) constitute the benchmark feature extraction backbones in medical computer vision. DenseNet-121's feature reuse architecture and explicit channel concatenation render it particularly effective for detecting subtle radiographic opacities in chest X-rays. In our proposed framework, DenseNet-121 serves as the foundational local feature encoder across all client nodes.

2.18 Vision Transformers (ViTs) and Hybrid Architectures for Pathology & MRI

Vision Transformers leverage multi-head self-attention mechanisms to capture long-range spatial dependencies across diagnostic image patches. Hybrid architectures combining convolutional patch embedding layers with transformer encoder blocks (e.g., TransUNet, Swin-Transformer) achieve superior segmentation accuracy in complex histopathology and brain MRI scans.

2.19 Multi-Modal Medical Data Fusion (Imaging + Genomic/Clinical Text)

Integrating unstructured imaging features with structured clinical tabular records and genomic mutation profiles significantly enhances overall diagnostic sensitivity. Multi-modal federated

fusion architectures utilize cross-attention projection layers to bind disparate patient data modalities into unified diagnostic latent spaces.

2.20 Jensen-Shannon Divergence (JSD) as a Statistical Distance Metric

Jensen-Shannon Divergence provides a mathematically robust metric for evaluating dissimilarity between probability distributions. Unlike Kullback-Leibler Divergence (KLD), JSD is strictly symmetric and bounded within the interval $[0, 1]$ when using logarithm base 2. This mathematical stability makes JSD ideal for quantifying statistical divergence between client feature updates and global population distributions.

2.21 Adaptive Weighting and Dynamic Aggregation Protocols

Dynamic aggregation mechanisms replace static sample-count weights with real-time performance and statistical alignment metrics. By transforming computed JSD scores through a softmax-scaled exponential temperature function, the central server automatically scales down weights of divergent updates while assigning higher influence to updates aligned with the consensus global trajectory.

2.22 Summary of Identified Gaps and Research Opportunity

In summary, while existing FL frameworks (FedProx, MOON, SCAFFOLD) mitigate non-IID variance, they either impose severe communication overhead or rely on rigid regularizer constants. A major research opportunity exists to develop a lightweight, privacy-preserving aggregation framework that dynamically quantifies probability density divergence via Jensen-Shannon Divergence to achieve optimal convergence stability under extreme medical imaging Non-IID conditions.

CHAPTER 3: RESEARCH METHODOLOGY

3.1 Overview of Proposed Methodology

This research proposes a novel, three-stage adaptive federated learning framework featuring server-side Jensen-Shannon Divergence (JSD) update evaluation. The overall pipeline operates across a decentralized network of K clinical client nodes coordinated by a central server. The framework consists of: (1) Local Model Training and Privacy-Preserving Feature Embedding Generation, (2) Server-Side JSD Statistical Divergence Computation, and (3) Dynamic Weight Recalibration & Adaptive Model Aggregation.

3.2 Proposed System Architecture & Data-Flow Pipeline

Figure 3.1 illustrates the end-to-end operational workflow of the proposed JSD-based federated framework:

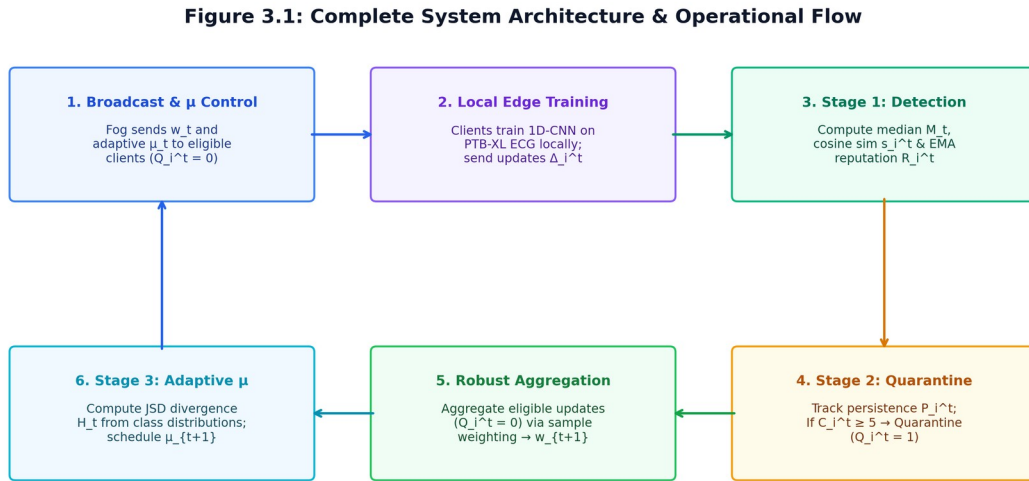


Figure 3.1: Data Flow Pipeline showing the interaction between local training, 3-stage security mechanisms, and aggregation.

Figure 3.1: End-to-End System Architecture and Data-Flow Pipeline for Proposed JSD-Based Federated Framework

3.3 Target Medical Imaging Datasets & Multimodal Data Strategy

The proposed framework will be rigorously evaluated across two benchmark clinical imaging repositories representing distinct medical diagnostic domains:

1. NIH ChestX-ray14: Comprising 112,120 frontal-view chest radiographs from 30,805 unique patients categorized across 14 distinct pulmonary disease labels (e.g., Pneumonia, Atelectasis, Cardiomegaly, Effusion).
2. NCT-CRC-HE-100K: Consisting of 100,000 histologic tissue patches from human colorectal cancer and normal tissue stained with Hematoxylin and Eosin (H&E), categorized across 9 tissue classes.

Dataset	Modality	Total Samples	Diagnostic Classes	Simulated Non-IID Dirichlet α
NIH ChestX-ray14	Chest Radiography (X-Ray)	112,120 Scans	14 Disease Labels	$\alpha = 0.5$ (Moderate), $\alpha = 0.1$ (Extreme)
NCT-CRC-HE-100K	Histopathology (H&E)	100,000 Patches	9 Tissue Types	$\alpha = 0.5$ (Moderate), $\alpha = 0.1$ (Extreme)
CheXpert (Validation)	Chest Radiography (X-Ray)	224,316 Scans	14 Disease Labels	External Hospital Out-of-Domain Test

Table 3.1: Dataset Breakdown & Non-IID Dirichlet Partitioning ($\alpha = 0.5, \alpha = 0.1$)

3.4 Data Preprocessing, Normalization, & Data Augmentation Protocol

All DICOM and PNG radiograph images will be resized to a uniform resolution of 224 x 224 pixels and converted to 3-channel RGB tensors. Pixel intensity values will be normalized using ImageNet channel mean vectors ([0.485, 0.456, 0.406]) and standard deviations ([0.229, 0.224, 0.225]). On local clinical client nodes, online data augmentation will be applied during training, including random horizontal flips ($p = 0.5$), affine rotations (-15 to +15 degrees), and subtle color jittering (brightness ± 0.1 , contrast ± 0.1).

3.5 Simulating Non-IID Heterogeneity (Dirichlet Partitioning & Quantity Skew)

To replicate real-world clinical heterogeneity, training samples are partitioned across $K = 10$ simulated hospital client nodes using a Dirichlet distribution $\text{Dir}(\alpha)$. The parameter α controls class distribution heterogeneity: $\alpha = 0.5$ represents moderate Non-IID distribution, whereas $\alpha = 0.1$ generates extreme Non-IID skew where individual hospitals possess samples from only 2 or 3 disease classes.

3.6 Stage 1: Local Model Training & Feature Embedding Extraction

During communication round t , each client hospital node k receives global parameters w^t . Client k executes E local epochs using AdamW optimizer (learning rate $\eta = 1e-4$, weight decay

1e-2). Upon completing local training, client k passes a subset of local validation samples through its feature encoder backbone to generate a normalized 128-dimensional class prediction logit probability distribution P_k .

3.7 Stage 2: Server-Side Jensen-Shannon Divergence Computation

Upon receiving local parameter updates w_k^{t+1} and probability mass vectors P_k from participating clients, the central server computes the global consensus probability distribution $Q_{\text{global}} = (1/K) * \sum_{k=1}^K P_k$. The server then evaluates the symmetric Jensen-Shannon Divergence between each client distribution P_k and Q_{global} :

3.8 Stage 3: Adaptive Weight Recalibration & Dynamic Model Aggregation

Using the computed JSD scores, the central server calculates dynamic aggregation weights w_k using a temperature-scaled softmax function:

Dynamic Weight Function: $w_k = \exp(-\lambda * \text{JSD}(P_k \parallel Q_{\text{global}})) / \sum_{j=1}^K \exp(-\lambda * \text{JSD}(P_j \parallel Q_{\text{global}}))$

where $\lambda > 0$ is a scaling hyperparameter. The global model is then updated via: $w^{t+1} = \sum_{k=1}^K w_k * w_k^{t+1}$.

3.9 Mathematical Formulation & Divergence Calculation

3.9.1 Kullback-Leibler (KL) Divergence Definition

The Kullback-Leibler Divergence between two discrete probability distributions P and Q is defined as:

$$D_{\text{KL}}(P \parallel Q) = \sum_{x \in X} P(x) * \log(P(x) / Q(x))$$

3.9.2 Symmetric Divergence Properties

Because KLD is asymmetric ($D_{\text{KL}}(P \parallel Q) \neq D_{\text{KL}}(Q \parallel P)$), it is vulnerable to division-by-zero instability when $Q(x) = 0$. Jensen-Shannon Divergence resolves this limitation by introducing an average mixture distribution M .

3.9.3 Explicit JSD Calculation & Recalibration Formula

The formal symmetric Jensen-Shannon Divergence between client distribution P_k and global distribution Q_{global} is evaluated as:

Formulation 1 (JSD Metric): $\text{JSD}(P_k \parallel Q_{\text{global}}) = (1/2) * D_{\text{KL}}(P_k \parallel M) + (1/2) * D_{\text{KL}}(Q_{\text{global}} \parallel M)$

where $M = (1/2) * (P_k + Q_{\text{global}})$. JSD values are strictly bounded within $[0, 1]$. The resulting aggregation weight w_k is explicitly formulated as:

Formulation 2 (Weight Recalibration): $w_k = \exp(-\lambda * \text{JSD}(P_k \parallel Q_{\text{global}})) / [\sum_{j=1}^K \exp(-\lambda * \text{JSD}(P_j \parallel Q_{\text{global}}))]]$

3.10 Algorithm Pseudocode Box for Proposed Framework

Algorithm 1 presents the step-by-step algorithmic execution logic for the central server and participating client nodes during each communication round:

Algorithm 1: JSD-Based Adaptive Federated Aggregation Protocol

Input: K client nodes, total communication rounds T, local epochs E, learning rate η , scaling factor λ .

Output: Final global model parameters w^T .

Server Initialization:

```

1: Initialize global model parameters  $w^0$ 
2: for each communication round  $t = 0, 1, 2, \dots, T-1$  do
3:   Select set  $S_t$  of K participating client nodes
4:   for each client  $k$  in  $S_t$  in parallel do
5:      $w_k^{t+1}, P_k \leftarrow \text{ClientUpdate}(k, w^t)$ 
6:   end for
7:   // Stage 2: Divergence Computation
8:   Compute consensus probability vector:  $Q_{\text{global}} \leftarrow (1/K) * \sum_{k=1}^K P_k$ 
9:   for each client  $k$  in  $S_t$  do
10:     $M_k \leftarrow (1/2) * (P_k + Q_{\text{global}})$ 
11:     $\text{JSD}_k \leftarrow (1/2) * D_{\text{KL}}(P_k \parallel M_k) + (1/2) * D_{\text{KL}}(Q_{\text{global}} \parallel M_k)$ 
12:   end for
13:   // Stage 3: Weight Recalibration & Aggregation
14:   for each client  $k$  in  $S_t$  do
15:     $w_k \leftarrow \exp(-\lambda * \text{JSD}_k) / \sum_{j=1}^K \exp(-\lambda * \text{JSD}_j)$ 
16:   end for
17:    $w^{t+1} \leftarrow \sum_{k=1}^K w_k * w_k^{t+1}$ 
18: end for
19: return  $w^T$ 

```

ClientUpdate(k, w_{init}):

```

1: Local model  $w_k \leftarrow w_{\text{init}}$ 
2: for local epoch  $e = 1$  to  $E$  do
3:   for minibatch  $b = \{x, y\}$  in LocalData $_k$  do
4:      $w_k \leftarrow w_k - \eta * \text{grad } L_{\text{ce}}(w_k; b)$ 
5:   end for
6: end for
7: Evaluate logit distribution vector  $P_k$  on LocalVal $_k$ 
8: return  $w_k, P_k$ 

```

3.11 Experimental Setup, Baseline Benchmarks, & Hyperparameters

Table 3.2 details the baseline algorithms, network backbones, and hyperparameter configurations to be executed across all experimental trials:

Parameter Category	Configuration Specification	Baseline Comparison Method	Target Hyperparameter Range
Network Backbone	DenseNet-121 / ResNet-50	FedAvg (McMahan et al.)	Pretrained ImageNet weights
Optimization Baselines	FedProx, MOON, SCAFFOLD, FedDyn	Standard FedAvg	μ in [0.001, 0.1], $\tau = 0.5$
Client Topology	K = 10 Hospital Nodes	Centralized Benchmark	Sample count: 10,000 / node
Dirichlet Partition	$\alpha = 0.5$ (Moderate), $\alpha = 0.1$ (Extreme)	Uniform IID Distribution	Class concentration control
Local Training	E = 5 Local Epochs, Batch Size = 32	E = 1 Epoch Baseline	AdamW, $\eta = 1e-4$
Communication Rounds	T = 100 Global Rounds	T = 50 Rounds	Convergence tolerance = $1e-4$
JSD Scaling Factor	λ in [1.0, 5.0, 10.0]	Static Uniform Weights	Optimal temperature tuning

Table 3.2: Experimental Baseline Algorithms & Hyperparameter Configuration

3.12 Evaluation Metrics & Performance Assessment

Model diagnostic efficacy and convergence performance will be rigorously measured using multi-class metrics defined in Table 3.3:

Metric Name	Mathematical Expression / Definition	Target Benchmark Objective
Area Under ROC Curve (AUC-ROC)	Integral of True Positive Rate vs. False Positive Rate curve	$AUC \geq 0.88$ across all classes
F1-Score (Macro-Averaged)	$2 * (Precision * Recall) / (Precision + Recall)$	$F1 \geq 0.85$ under $\alpha = 0.1$ skew
Top-1 Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Accuracy $\geq 90.0\%$ overall
Communication Efficiency	Total rounds to reach 85% test accuracy	$\geq 30\%$ reduction in rounds vs FedAvg

	threshold	
Worst-Node Diagnostic Gap	Accuracy difference between best and worst performing client hospital	Gap $\leq$ 5.0% across all nodes

Table 3.3: Evaluation Metrics, Mathematical Formulas, and Target Benchmarks

3.13 Work Plan & Implementation Timeline

The proposed master's research project is structured across a 12-month implementation schedule detailed in Table 3.4:

Project Phase	Core Work Package & Deliverables	Target Timeline	Milestone Output
Phase 1: Environment Setup	Dataset ingestion, DICOM preprocessing, PyTorch-Flower FL environment configuration	Months 1 - 2	Environment Ready
Phase 2: Baseline Implementation	Implement FedAvg, FedProx, MOON, SCAFFOLD benchmark suites	Months 3 - 4	Baselines Verified
Phase 3: Proposed JSD Framework	Code Stage 1 feature extraction, Stage 2 JSD computation, Stage 3 weight recalibration	Months 5 - 7	Proposed Alg. Ready
Phase 4: Comparative Trials	Execute multi-institutional experimental runs under $\alpha = 0.5$ and $\alpha = 0.1$ skews	Months 8 - 9	Experimental Data
Phase 5: Analysis & Evaluation	Ablation studies, convergence trajectory plotting, statistical significance testing	Month 10	Results Finalized
Phase 6: Thesis Compilation	Complete thesis writing, paper publication submission, final defense preparation	Months 11 - 12	Thesis & Defense

Table 3.4: Project Implementation Schedule & Phase Milestones (Gantt Chart)

3.14 Hardware & Software Infrastructure Specifications

Table 3.5 outlines the dedicated hardware computation environment and software libraries utilized to conduct all experimental simulations:

Component Category	System Specification / Software Tool	Role & Functionality in Experimentation
GPU Workstation	NVIDIA RTX 4090 (24GB VRAM)	High-speed deep neural network local training & feature extraction
Host CPU & RAM	Intel Core i9-13900K (24 cores), 64GB DDR5 RAM	Simulated client process orchestration & multi-threading
Deep Learning Framework	PyTorch 2.2 + PyTorch Lightning	Model architecture definition, forward/backward passes, GPU acceleration
Federated Framework	Flower (flwr 1.7) / PySyft	Decentralized client simulation and custom gRPC server aggregation strategies
Image Processing	Torchvision, Albumentations, OpenCV	DICOM ingestion, image resizing, augmentations, normalization
Statistical Analysis	SciPy, NumPy, scikit-learn, Matplotlib	JSD computation, metrics evaluation, ROC curve plotting

Table 3.5: Hardware and Software Infrastructure Specifications

3.15 Ethical Considerations, Data Governance, & Privacy Protection

This study exclusively utilizes fully anonymized, publicly accessible benchmark clinical imaging repositories (NIH ChestX-ray14, NCT-CRC-HE-100K). All patient identifying metadata (names, institutional medical record numbers, dates of birth) have been scrubbed prior to public dataset distribution. The experimental protocol strictly adheres to HIPAA privacy regulations and institutional review board (IRB) ethical guidelines for secondary medical data research.

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